Thus, $2x + 3x = 5x$ and $2y + 4y = 6y$.

However, $5x$ and $7y$ are **unlike terms** and cannot be added.

Variables

In the expression, $2x + 3y + 4$, we do not know the values of x and y.

Until we are given more information to find x and y, their values are **variable**.

Letters of the alphabet are used as variables e.g. x, y, a, b, c etc.

The number 4 does not change. Since it is fixed, we call it a **constant**.

In $2x$, the number 2 tells us how many xs there are.

We call 2 the **coefficient** of x. Similarly, 3 is the coefficient of y.

Notation

In algebra, the symbols $\times$ and $\div$ are not normally used.

Instead of writing $2 \times k$, we just write $2k$.

We write $\frac{a}{b}$ instead of $a \div b$.

Also
(i) $x + x + x = 3x$
(ii) $2 \times a \times b = 2ab$
(iii) $4 \times x \times y \times 3 = 4 \times 3 \times x \times y = 12xy$
(iv) $2 \times a \times b \times c = 2abc$

Remember:

x … variable
$3x$ … term
$3x - 2y + 5$ … expression
$3x, 6x$ … like terms
$3x, 6y$ … unlike terms
5 … constant
$3x$, 3 is the coefficient of x.
$1x$ is written as x
$1x + 1y$ is written as $x + y$.

Example 1

Simplify each of these by adding like terms:
(i) $x + 2x + 3y + 5y$ (ii) $3a + 4b - a + 2b + 2a$

(i) $x + 2x + 3y + 5y = 3x + 8y$

(ii) $3a + 4b - a + 2b + 2a = \underbrace{3a - a} + 2a + 4b + 2b$
$= 2a + 2a + 6b$
$= 4a + 6b$

Exercise 8.2

1. Write each of the following without using the multiplication sign:
(i) $5 \times a$ (ii) $4 \times b$ (iii) $4 \times 2 \times x$ (iv) $3 \times 4 \times y$
(v) $4 \times 2c$ (vi) $5a \times 3$ (vii) $a \times b$ (viii) $4 \times a \times b$

2. Write these without the multiplication sign:

(i) $2a \times 4b$ (ii) $3x \times 7y$ (iii) $3 \times ab$ (iv) $a \times b \times c$
(v) $3xy \times 2$ (vi) $ab \times c$ (vii) $3xy \times 4z$ (viii) $2ab \times 5c$

3. Simplify each of the following by adding like terms:

(i) $x + x + x$ (ii) $x + 2x + 3x$ (iii) $a + 2a + 3 + 4$
(iv) $2b + 4 + 5b + 2$ (v) $3x + 4 + 2x + 6$ (vi) $x + y + 3x + 2y$

4. Simplify these by adding like terms:

(i) $5x - 2x$ (ii) $3a - a$ (iii) $5x - 2x$
(iv) $3a + 2 - a + 5$ (v) $6b - 2 + b - 4$ (vi) $3a - 2b + 2a + 4b$

5. Simplify each of these:

(i) $12a + b + 3a + 5b$ (ii) $3x + 2y + 3 + 4x + 3y + 1$
(iii) $5x - 4 + 2x + 8$ (iv) $7x - 4 - 3x + 7$
(v) $6a + b + 3 + 2a + 2b - 1$ (vi) $3x + 4 + 2x - 6 + x + 3$

6. Simplify each of these expressions:

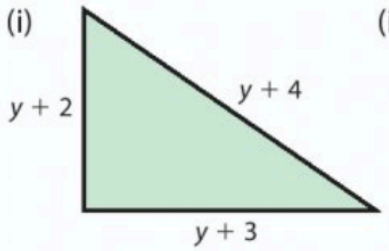
(i) $ab + ab$ (ii) $3ab + 2ab$ (iii) $4xy - 2xy$
(iv) $4cd + 5cd - 2cd$ (v) $2ab - 5ab + 4ab$ (vi) $5xy + 6xy - 7xy$

7. Simplify these expressions:

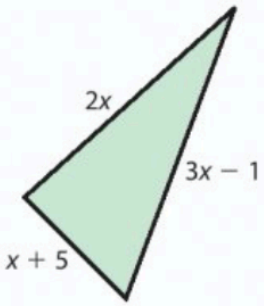
(i) $2p + 3q - r + p - 4q + 2r$ (ii) $5k + 3 - 4k + 6 + k - 4$
(iii) $2ab + c + 5ab - 4c$ (iv) $3xy + 2z + xy + 9z$
(v) $6ab + 2cd - ab + 3cd$ (vi) $6x - xy + 5x - 7xy$

8. Find and simplify an expression for the perimeter (length all round) of each of the following figures:

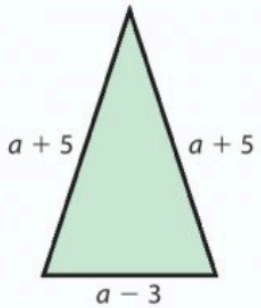
(i)



(ii)

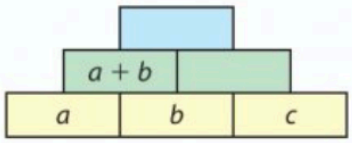


(iii)

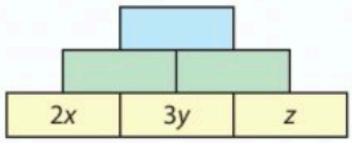


9. The number in each box is found by adding the numbers in the two boxes below it. Write an expression for the blue box as simply as possible.

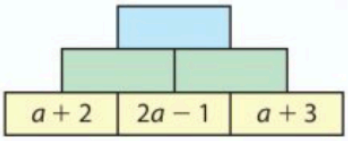
(i)



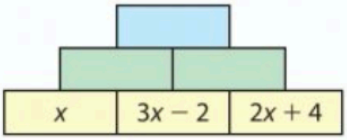
(ii)



(iii)



(iv)



10. Here are six cards containing expressions:

(i) Which two cards total $5x - 5$?
(ii) Which two cards total $2x + 8$?
(iii) Which two cards total $x + 3$?
(iv) Which two cards total $10 - 3x$?
(v) What is the total of all the expressions on the cards?

11. $3x + 5y - 8$ is an expression.
(i) Name the two variables in this expression.
(ii) Why are they called variables?
(iii) Write down the coefficient of x.
(iv) Write down the constant in the expression.
(v) Are there any like terms in the expression?

12. An expression contains three terms.
The variable in one term is c; the variable in another term is d; there is also a term which is a constant.
What is the expression if
(i) the coefficient of c is 4
(ii) the coefficient of d is -3
(iii) the constant term is a prime number between 6 and 10 ?